

Monochromator Testing Report

Project No.:	17125
End User:	University of California, Santa Cruz
End User Country:	United States

Manufacturer:	Sciencetech Inc.
Model:	9055
Motor ID:	Trinamic
System Voltage Specification:	24 VDC
Serial No.:	0004036

Performed By:	JF
Date [YYYY-MM-DD]:	2024-09-19
Reviewed By:	RP
Date of Issue [YYYY-MM-DD]:	2024-09-23

Note

Tests are performed using cosine correcting optics or sources with uniform and wide angular distribution [such as a Hg pen lamp] at the input aperture of the monochromator. Calibration is only guaranteed when coupling similar sources or using cosine corrected optics.

Coupling of sources to the monochromator without using cosine corrected optics may result in unexpected spectral line shifts due to angular misalignment of the external source. In these cases, calibration can usually be corrected by performing a shift of the zero-order location.

Repeatability

The repeatability test is a test of wavelength repeatability. The test is designed to simulate real working conditions where gratings are changed and the turret is moved through a series of sequences. Tests are performed with each grating. The slit width and step size used provides a spectral resolution that allows for measuring the repeatability specification. The test is performed on a single spectral line within the grating spectral range [nm].

Process

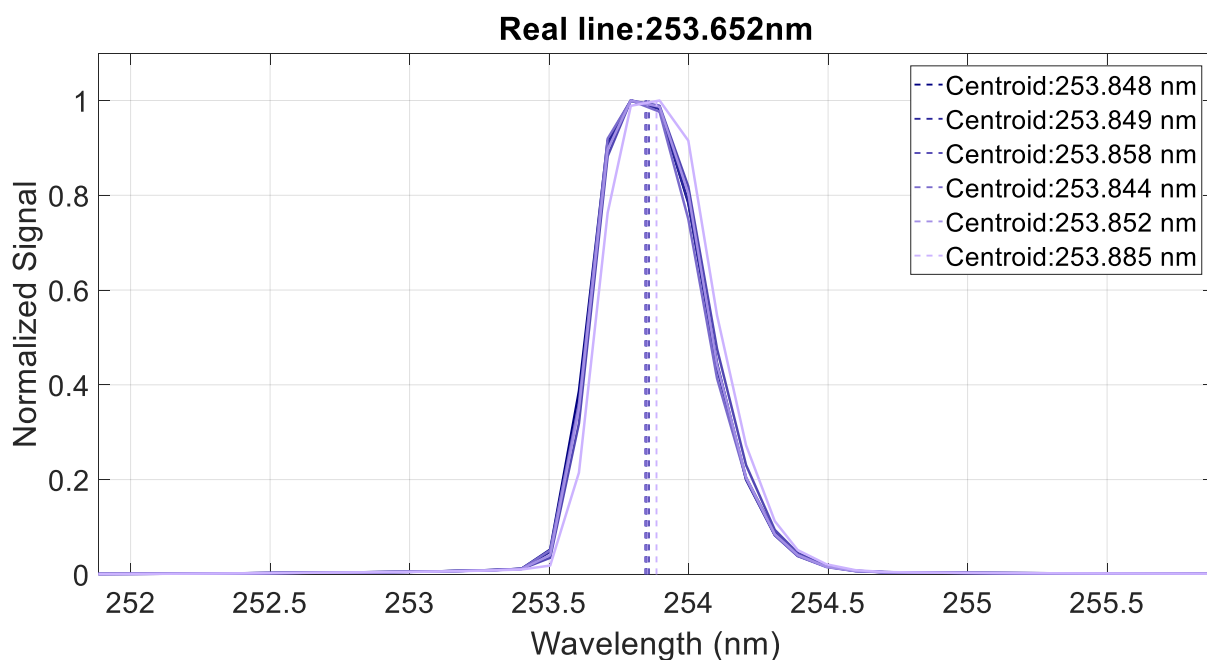
1. Home the monochromator.
2. Measure the spectral line and determine location.
3. Change grating if multi-grating system.
4. Change back to grating under test if multi-grating system.
5. Move to a new wavelength position.
6. Scan the spectral line under test and determine location.

Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	1200 lines/mm @ 300 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.1
Slit Height [mm]:	1
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	253.652	253.848	-0.196
2	253.652	253.849	-0.197
3	253.652	253.858	-0.206
4	253.652	253.844	-0.192
5	253.652	253.852	-0.200
6	253.652	253.885	-0.233
Average Difference [nm]:		-0.204	
Measured Repeatability [$\pm$ nm]:		0.021	
Repeatability Specification ¹ [$\pm$ nm]:		0.100	
Pass/Fail ¹ :		PASS	

1. Quoted repeatability range is for a 1200l/mm grating near the blazed wavelength.

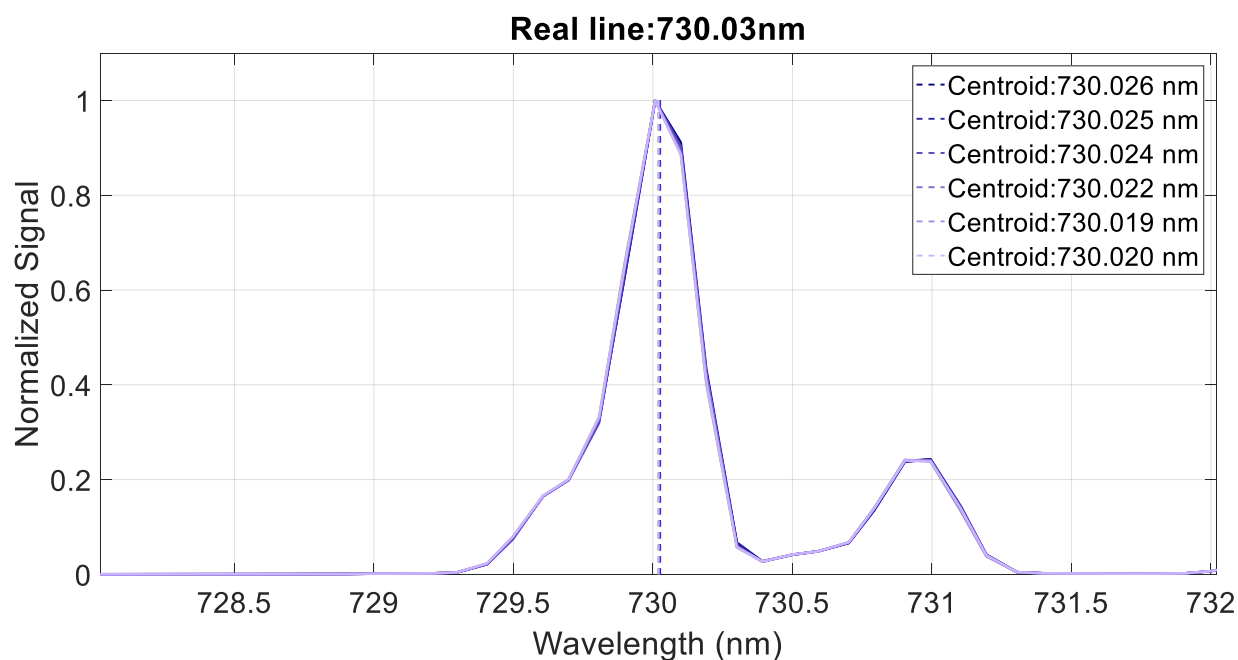


Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	1200 lines/mm @ 750 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.1
Slit Height [mm]:	1
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	730.030	730.026	0.004
2	730.030	730.025	0.005
3	730.030	730.024	0.006
4	730.030	730.022	0.008
5	730.030	730.019	0.011
6	730.030	730.020	0.010
Average Difference [nm]:		0.007	
Measured Repeatability [$\pm$ nm]:		0.003	
Repeatability Specification ¹ [$\pm$ nm]:		0.100	
Pass/Fail ² :		PASS	

1. Quoted repeatability range is for a 1200/mm grating near the blazed wavelength.

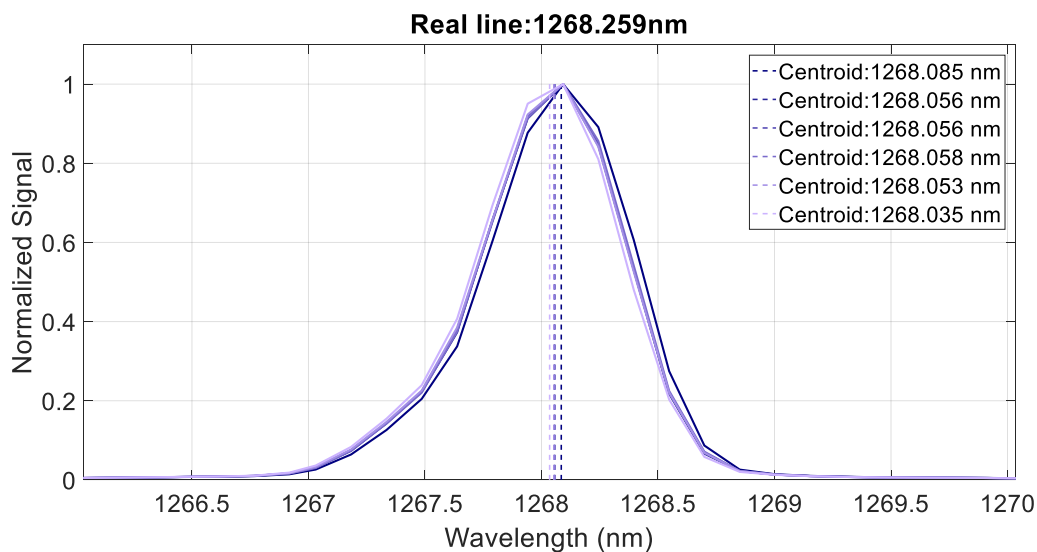


Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	600 lines/mm @ 1250 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.15
Slit Height [mm]:	1
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	1268.259	1268.085	0.174
2	1268.259	1268.056	0.203
3	1268.259	1268.056	0.203
4	1268.259	1268.058	0.201
5	1268.259	1268.053	0.206
6	1268.259	1268.035	0.224
Average Difference [nm]:		0.202	
Measured Repeatability [$\pm$ nm]:		0.025	
Repeatability Specification ¹ [$\pm$ nm]:		0.100	
Pass/Fail ² :		PASS	

1. Quoted repeatability range is for a 1200l/mm grating near the blazed wavelength.



Spectral Calibration Accuracy

The spectral calibration accuracy test is designed to measure the accuracy and reproducibility of the spectral calibration. Tests are performed with each grating. The slit width and step size used provides a spectral resolution that allows for measuring the spectral calibration accuracy specification. The test is performed on a at least 3 spectral lines covering the majority of the grating's spectral range [nm].

Process

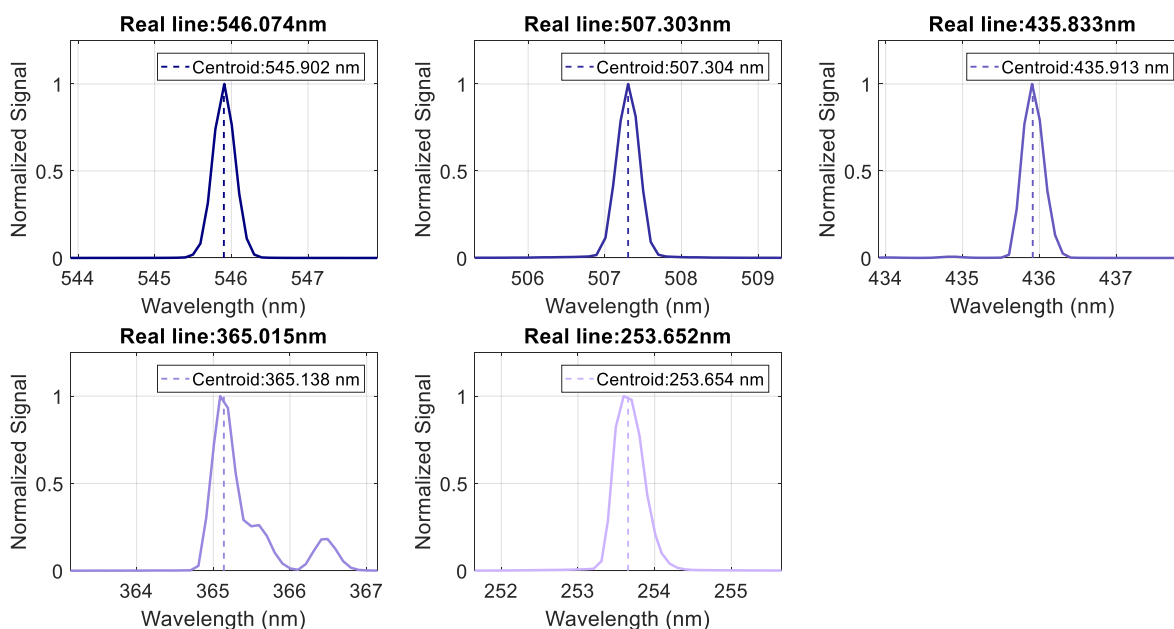
1. Home the monochromator.
2. Measure the spectral range [nm] of interest.
3. Analyze the locations of the spectral lines measured.

Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	1200 lines/mm @ 300 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.1
Slit Height [mm]:	1
Range [nm]:	250 - 549
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	253.652	253.654	-0.002
2	365.015	365.138	-0.123
3	435.833	435.913	-0.080
4	507.303	507.304	-0.001
5	546.074	545.902	0.172
Average Difference [nm]:		-0.007	
Measured Accuracy [$\pm$ nm]:		0.148	
Accuracy Specification ¹ [$\pm$ nm]:		0.200	
Pass/Fail ² :		PASS	

1. Quoted accuracy range is for a 1200/mm grating near the blazed wavelength.

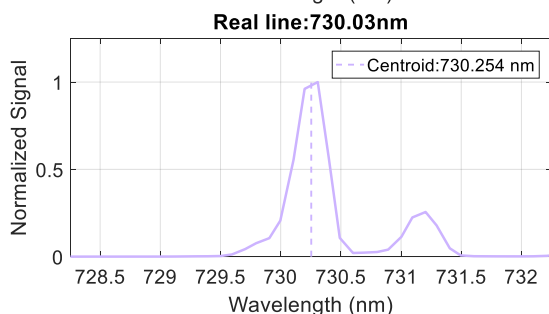
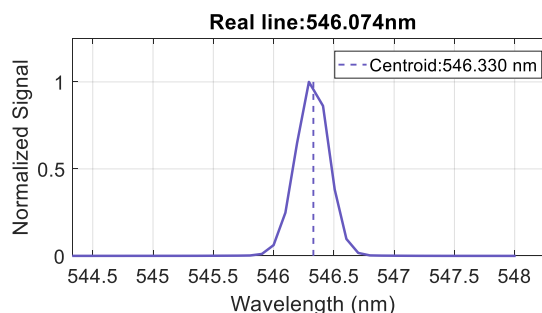
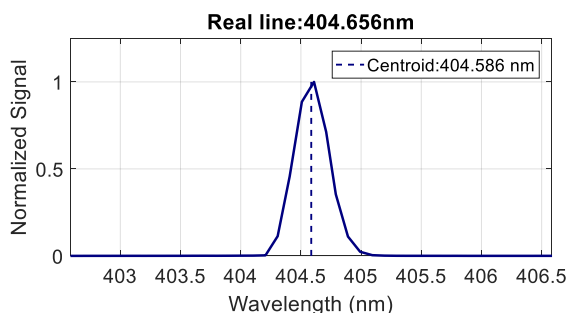


Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	1200 lines/mm @ 750 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.1
Slit Height [mm]:	1
Range [nm]:	401 – 733
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	404.656	404.586	0.07
3	546.074	546.330	-0.256
4	730.030	730.254	-0.224
Average Difference [nm]:		-0.137	
Measured Accuracy [$\pm$ nm]:		0.163	
Accuracy Specification ¹ [$\pm$ nm]:		0.200	
Pass/Fail ¹ :		PASS	

1. Quoted accuracy range is for a 1200/mm grating near the blazed wavelength.

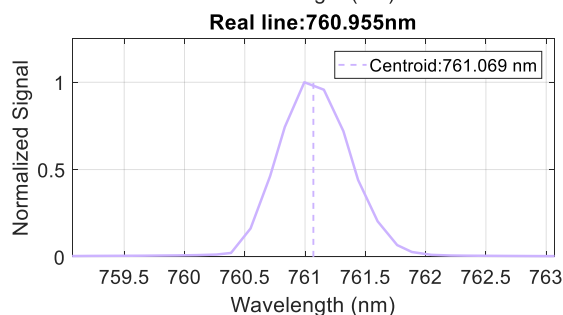
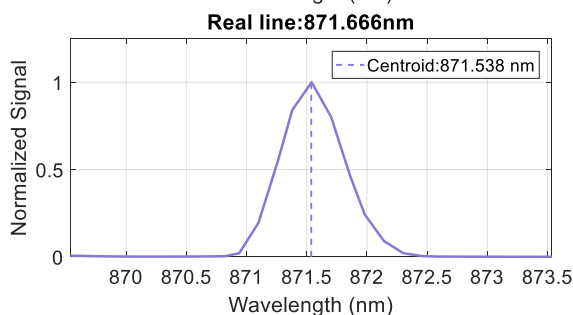
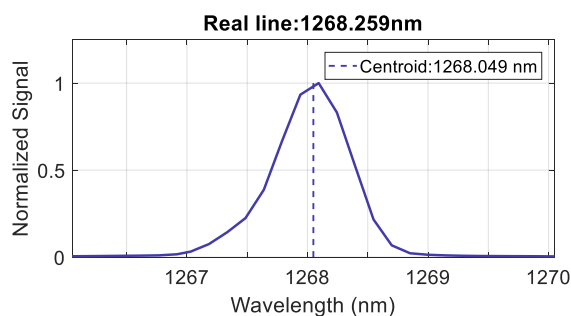
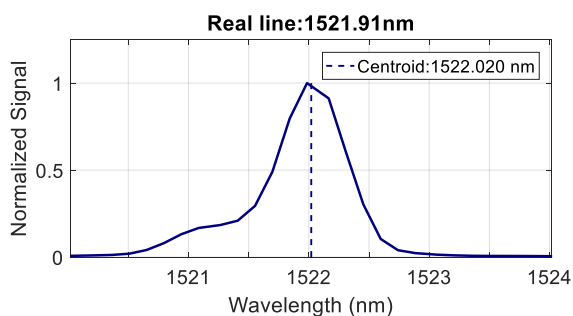


Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	600 lines/mm @ 1250 nm
Step Size [nm]:	0.1
Slit Width [mm]:	0.15
Slit Height [mm]:	1
Range [nm]:	765 - 1273
Source:	Hg Pen Lamp
Diffuser:	None

Test No.	Real Line Location [nm]	Measured Line Location [nm]	Difference [nm]
1	760.955	761.069	-0.114
2	871.666	871.538	0.128
3	1268.259	1268.049	0.210
4	1521.910	1522.020	-0.110
Average Difference [nm]:		0.028	
Measured Accuracy [$\pm$ nm]:		0.162	
Accuracy Specification ¹ [$\pm$ nm]:		0.200	
Pass/Fail ¹ :		PASS	

1. Quoted accuracy range is for a 1200l/mm grating near the blazed wavelength.



Resolution Test

The resolution test measures the capability of the monochromator to resolve a single wavelength. The result is the spectral line full width half max [FWHM]. Tests are performed with one grating. The slit width and step size used provides a spectral resolution that allows for measuring the spectral resolution specification. The test is performed on at zero order and one spectral line.

Process

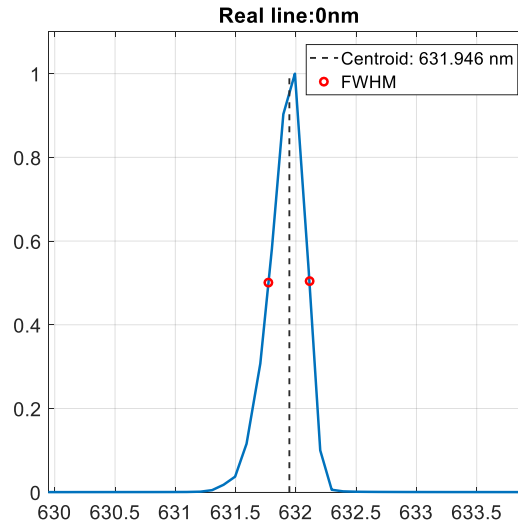
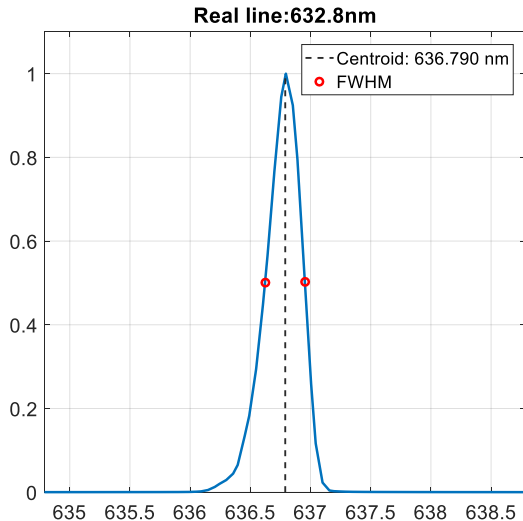
1. Home the monochromator.
2. Measure the HeNe laser line.
3. Analyze the FWHM.

Results

Measurement Configuration	
Monochromator:	9055
Input Location:	Side
Output Location:	Side
Grating Pitch and Blaze:	1200 lines/mm @ 300 nm
Step Size [nm]:	0.05
Slit Width [mm]:	0.10
Slit Height [mm]:	1
Source:	HeNe Laser
Diffuser:	Diffusing Tape

FWHM at 632.8 nm ¹ [nm]:	0.33
FWHM at 0 nm ² [nm]:	0.34
FWHM Specification ³ [nm]:	0.40
Pass/Fail:	PASS

- Resolution measurements are performed before spectral calibration.
- 0 nm is the configuration in which there is total specular reflection from the grating surface.
- Resolution measured at 0.1 mm slit width. The specified resolution under these conditions is 0.4 nm FWHM. Quoted resolution of system is at 0.05 mm slit width and 1200l/mm grating near the blaze wavelength.



Light Tightness Test

This test measures the light tightness of the system.

Process

A light source is placed around all seams and corners of the monochromator body while a PMT detector attached to the monochromator output is monitored for any increase in signal.

Results

Monochromator:	9055
Input Slit Width [mm]:	0.1
Output Slit Width [mm]:	0.1
Detector:	PMT, R928 Tube
Gain:	10 X
HV:	900 VDC
Test Light Source:	White LED
Pass/Fail:	PASS